

Photon-Path Spectral Fitting

Draft Main-Text and Appendix Methods

OCO-2 cloud-proximity analysis

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This standalone document contains the proposed main-text Methods subsection and supporting appendix. The source fragment is designed to be inserted directly into the eventual manuscript.

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0.1 Spectrum-internal photon-path features

We derived spectral features that summarize the distribution of atmospheric photon paths represented in each OCO-2 absorption band. The derivation follows the radiative-transfer equivalence theorem (Irvine, 1964; Partain et al., 2000; Stephens and Heidinger, 2000). For a fixed illumination and viewing geometry, let $P_0(\Gamma)$ denote the ensemble of photon trajectories Γ reaching the detector in a conservative reference atmosphere with the same surface boundary and continuum scattering properties as the absorbing atmosphere. Across each narrow OCO-2 band, we assume that these scattering properties vary slowly relative to molecular line absorption. Wavelength then changes the contribution of a trajectory primarily through its Beer–Lambert survival probability,

$$w_\lambda(\Gamma) = \exp[-A_\lambda(\Gamma)], \quad A_\lambda(\Gamma) = \int_\Gamma \alpha_\lambda(\mathbf{r}) \, ds, \quad (1)$$

where α_λ is the local molecular absorption coefficient. The normalized radiance is therefore the conservative-ensemble mean

$$\mathcal{T}_\lambda = \int P_0(\Gamma) \exp[-A_\lambda(\Gamma)] \, d\Gamma. \quad (2)$$

This construction separates the multiple-scattering calculation from molecular absorption and remains applicable to three-dimensional cloud fields because the cloud geometry is encoded in $P_0(\Gamma)$ (Stephens and Heidinger, 2000).

To obtain a tractable scalar representation, we approximate the absorber amount along each trajectory as

$$A_\lambda(\Gamma) \simeq \tau_\lambda \ell'(\Gamma), \quad (3)$$

where τ_λ is the modeled molecular optical depth along the direct geometrical slant path and ℓ' is an effective absorber-path enhancement relative to that reference. The conservative ensemble induces a normalized distribution $p_0(\ell')$, so Eq. (2) reduces to

$$\mathcal{T}(\tau_\lambda) = \int_0^\infty p_0(\ell') \exp(-\tau_\lambda \ell') \, d\ell'. \quad (4)$$

Thus, the resolved absorption spectrum samples the Laplace transform of one wavelength-independent conservative path distribution within each band. This does not imply that weak and strong channels detect identical photon populations: absorption reweights the detected paths as $p_{\text{det}}(\ell' \mid \tau) \propto p_0(\ell') \exp(-\tau \ell')$, preferentially suppressing long paths at large τ .

We normalized the measured radiance as

$$T_\lambda = \frac{\pi I_\lambda}{F_{0,\lambda} \mu_0} = \gamma \mathcal{T}(\tau_\lambda), \quad (5)$$

where $F_{0,\lambda}$ is the top-of-atmosphere solar irradiance, μ_0 is the cosine of the solar zenith angle, and γ is an effective continuum reflectance. In practice, γ can include surface bidirectional reflectance, polarization, and surface–atmosphere coupling and is not interpreted as an exact Lambertian albedo.

Let C_n denote the ordinary cumulants of $p_0(\ell')$. Expanding the log Laplace transform about zero absorption gives

$$\ln T(\tau) = \ln \gamma + \sum_{n=1}^{\infty} \frac{(-1)^n C_n}{n!} \tau^n. \quad (6)$$

For each sounding and band, we fitted the truncated form

$$\ln T(\tau) \simeq b + \sum_{n=1}^N \frac{(-1)^n k_n}{n} \tau^n, \quad k_n = \frac{C_n}{(n-1)!}, \quad (7)$$

where b is the fitted continuum intercept. Under the equivalence-theorem and scalar-path assumptions, $k_1 = E_{p_0}[\ell']$ is the mean absorber-path enhancement and $k_2 = \text{var}_{p_0}(\ell')$ is its variance; coefficients $k_{n \geq 3}$ are scaled higher cumulants.

The model is linear in $(b, k_1, \dots, k_N)$. We solved it by singular-value least squares and used bounded-variable least squares only when the unconstrained solution violated $k_1, k_2 \geq 0$. We excluded the two edge channels and channels with $T_\lambda > 1$. The truncation orders were 7, 3, and 7 for the O₂ A, weak-CO₂, and strong-CO₂ bands, respectively; the lower weak-CO₂ order reflects its smaller optical-depth range and the resulting poor identifiability of high-order terms. Fits were produced for both raw and Savitzky–Golay-smoothed log transmittance; the raw coefficients were used in the production model, while the smoothed coefficients were retained for a robustness comparison.

We interpret these quantities as *band-effective absorber-path cumulants*. The scalar approximation in Eq. (3) can fail when pressure- and temperature-dependent line absorption weights atmospheric layers differently, or when surface, aerosol, or polarization properties vary within a band. A layer-resolved formulation and the distinction from a gamma path model are given in Appendix 1.

1 Photon-path derivation, coefficient convention, and limitations

1.1 Equivalence theorem and conservative path ensemble

The equivalence theorem separates the statistics of photon transport from the spectral absorption applied along those paths (Irvine, 1964). In its most general form relevant here, the normalized radiance is

$$\mathcal{T}_\lambda = E_{P_0} \{ \exp[-A_\lambda(\Gamma)] \}, \quad A_\lambda(\Gamma) = \int_\Gamma \alpha_\lambda(\mathbf{r}) \, ds. \quad (8)$$

The theorem requires absorption to modify photon weights without changing the geometry, scattering phase function, or boundary conditions that generate $P_0(\Gamma)$. It does not require a plane-parallel medium.

Stephens and Heidinger (2000) used this construction for molecular-line absorption in heterogeneous clouds. A continuum Monte Carlo calculation supplied a photon optical-path distribution and continuum radiance at each spatial location; an entire high-resolution O₂ A-band spectrum was then reconstructed by applying the wavelength-dependent transmission function to those stored path statistics. Direct multiple-scattering and path-distribution reconstructions agreed in their test, while changes between plane-parallel and heterogeneous clouds appeared through changes in the path distribution itself. Their O₂ A-band application subsequently demonstrated sensitivity to cloud optical depth, phase function, cloud pressure and pressure thickness, and surface albedo (Heidinger and Stephens, 2000).

Partain et al. (2000) generalized the computational use of the theorem by retaining geometrical path and scattering statistics that allow spectral gas, particle, and surface absorption to be applied after the multiple-scattering calculation. Their treatment accommodates multiple homogeneous layers and vertical gas profiles. The full joint distribution is memory-intensive, however, and reduced-PDF approximations discard trajectory information. The scalar distribution fitted here is deliberately simpler than that generalized bookkeeping: it retains one effective absorber-path coordinate per band rather than the complete layer-resolved trajectory history.

1.2 Scalar path reduction and detected-path selection

Let $\bar{L}_z$ be the direct geometrical path through atmospheric layer z , and let $L_z(\Gamma)$ be the corresponding path accumulated by trajectory Γ . The exact layered absorber amount is

$$A_\lambda(\Gamma) = \sum_z \alpha_{\lambda z} L_z(\Gamma). \quad (9)$$

The one-dimensional model assumes approximately proportional layer paths,

$$L_z(\Gamma) \simeq \ell'(\Gamma) \bar{L}_z, \quad (10)$$

which yields

$$A_\lambda(\Gamma) \simeq \ell'(\Gamma) \sum_z \alpha_{\lambda z} \bar{L}_z = \tau_\lambda \ell'(\Gamma). \quad (11)$$

The conservative trajectory ensemble then induces

$$p_0(\ell') = \int P_0(\Gamma) \delta[\ell' - \ell'(\Gamma)] d\Gamma, \quad (12)$$

and Eq. (8) becomes the one-dimensional Laplace transform

$$\mathcal{T}(\tau) = \int_0^\infty p_0(\ell') e^{-\tau \ell'} d\ell'. \quad (13)$$

The reference $\ell' = 1$ represents the modeled direct surface-slant absorber path. Multiple scattering can give $\ell' > 1$, whereas reflection above the surface can truncate the path and give $0 \leq \ell' < 1$. Accordingly, the fit requires nonnegative, not necessarily greater-than-unity, mean path enhancement.

The distribution among photons detected at a finite optical depth is

$$p_{\text{det}}(\ell' | \tau) = \frac{p_0(\ell') e^{-\tau \ell'}}{\mathcal{T}(\tau)}. \quad (14)$$

It follows that

$$-\frac{d \ln \mathcal{T}}{d\tau} = E_{\text{det},\tau}[\ell'], \quad (15)$$

$$\frac{d^2 \ln \mathcal{T}}{d\tau^2} = \text{var}_{\text{det},\tau}(\ell'). \quad (16)$$

Weak and strong channels therefore observe different absorption-selected path populations even when they sample one common conservative $p_0(\ell')$.

High-resolution O₂ A-band measurements provide an observational basis for this representation. Min et al. (2004) treated the resolved line spectrum as an inverse-Laplace problem and emphasized the requirements imposed by spectral resolution, slit-function characterization, stray light, and out-of-band rejection. Their O₂ and nearby water-vapor results also showed that different absorber profiles yield different effective path moments despite similar scattering wavelengths. Scholl et al. (2006) represented paths above, within, and below cloud separately, used pressure- and temperature-dependent O₂ absorption, and convolved the modeled spectrum with the measured slit function. Their analysis concentrated on the first two moments and generally used a gamma path PDF. These studies motivate the present moment-based fit but also show that its scalar path is an absorber-weighted approximation, not a complete reconstruction of geometrical trajectories.

1.3 Cumulant convention used by the estimator

Let C_n be the ordinary cumulants of $p_0(\ell')$. Since $\mathcal{T}(\tau)$ is the moment-generating function evaluated at $-\tau$,

$$\ln \mathcal{T}(\tau) = \sum_{n=1}^{\infty} \frac{(-1)^n C_n}{n!} \tau^n. \quad (17)$$

Including the effective continuum factor gives

$$\ln T(\tau) = \ln \gamma + \sum_{n=1}^{\infty} \frac{(-1)^n C_n}{n!} \tau^n. \quad (18)$$

The estimator is parameterized as

$$\ln T(\tau) \simeq b + \sum_{n=1}^N \frac{(-1)^n k_n}{n} \tau^n, \quad (19)$$

and therefore

$$b = \ln \gamma, \quad k_n = \frac{C_n}{(n-1)!}. \quad (20)$$

Thus $k_1 = C_1$ and $k_2 = C_2$ are exactly the mean and variance, whereas $k_3 = C_3/2!$, $k_4 = C_4/3!$, and higher k_n are scaled cumulants. This factorial convention follows from the $1/n$ factors used in the design matrix and should be retained when comparing the fitted coefficients with standard probability notation.

The finite-order model uses the design matrix

$$\mathbf{X}(\tau) = \left[1, -\tau, \frac{\tau^2}{2}, -\frac{\tau^3}{3}, \dots, \frac{(-1)^N \tau^N}{N} \right]. \quad (21)$$

Only k_1 and k_2 are constrained nonnegative. This is necessary for their mean and variance interpretation, while higher cumulants of a general distribution may have either sign. These local constraints do not guarantee that the complete fitted polynomial is the log Laplace transform of a globally nonnegative PDF.

1.4 Gamma distribution as a diagnostic special case

For comparison with classical path-distribution retrievals (Funk and Pfeilsticker, 2003; Scholl et al., 2006), suppose $p_0(\ell')$ is gamma distributed with mean m and shape κ :

$$p_0(\ell') = \frac{(\kappa/m)^\kappa}{\Gamma(\kappa)} \ell'^{\kappa-1} \exp\left(-\frac{\kappa}{m} \ell'\right). \quad (22)$$

Its Laplace transform is

$$T(\tau) = \gamma \left(1 + \frac{m}{\kappa} \tau\right)^{-\kappa}. \quad (23)$$

The ordinary cumulants are $C_n = (n-1)! m^n / \kappa^{n-1}$, giving

$$k_n = \frac{m^n}{\kappa^{n-1}}, \quad \kappa = \frac{k_1^2}{k_2}, \quad k_n = k_1 \left(\frac{k_2}{k_1}\right)^{n-1}. \quad (24)$$

The production estimator does not impose Eq. (23) or the coefficient sequence in Eq. (24); it fits k_n independently. Consequently, k_1^2/k_2 is a gamma-equivalent shape diagnostic rather than evidence that the inferred path distribution is gamma. A gamma PDF cannot generally represent a direct-beam-plus-scattered-tail mixture or distinct paths associated with multiple cloud layers (Bennartz et al., 2006).

1.5 Layer-resolved generalization and interpretation limits

Without Eq. (10), the appropriate conservative-ensemble expression is the multidimensional transform

$$T_\lambda = \gamma_\lambda \int p_0(L_1, \dots, L_Z) \exp \left[- \sum_z \alpha_{\lambda z} L_z \right] dL_1 \cdots dL_Z. \quad (25)$$

Different wavelengths sample different directions in the vector of layer-dependent absorption coefficients. The one-dimensional model can fail when cloud-top reflection removes below-cloud paths, three-dimensional transport adds path preferentially near cloud altitude, aerosols alter lower-tropospheric paths, or line centers and wings have materially different vertical absorption kernels. Wavelength-dependent surface bidirectional reflectance, polarization, and aerosol scattering can additionally violate the common conservative-ensemble assumption.

These effects do not invalidate the trajectory-level equivalence theorem. They limit the reduction from Eq. (8) to Eq. (13). We therefore use k_1 and k_2 as band-effective absorber-path cumulants. Exact identification with geometrical path moments would require closure against layer-resolved Monte Carlo photon histories. Useful closure tests include predicting strong channels after fitting weak and intermediate optical depths, comparing channels with similar total τ but different vertical kernels, and fitting synthetic spectra for which the conservative and layer-resolved path moments are known.

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